

COMPOSTING EDITION · PRACTICAL GUIDEBOOK



COMPOSTING BASICS

Reclaiming the Soil Gold

A comprehensive master manual on aerobic thermal composting, carbon-nitrogen thermodynamics, microbial activation, and field regeneration for Indian agriculture.

JAIVIK SATHI HANDBOOK · EDITION VI

PROSPERITY THROUGH LIVING SOIL & SUSTAINABLE TRADITION · POWERED BY AGENT FARMER

SYLLABUS & INDEX

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A Note to Indian Farmers: This handbook is designed as a field-ready reference. Each chapter pairs biological principles with practical, low-cost steps suitable for village farmsteads across all agro-climatic zones.

CHAPTER 01

The Crisis of the Flame

 *Why burning crop residue destroys the living heartbeat of Indian soils*

Every winter, the horizons of Northern India glow with a destructive orange fire. Millions of tonnes of paddy straw, wheat stubble, and sugarcane bagasse are set ablaze, sending thick, choking smog across thousands of villages and cities.

This is not merely an air quality crisis—it is an **agricultural catastrophe**. In those raging flames, billions of rupees worth of vital nitrogen, phosphorus, potash, and active organic carbon are vaporized into the sky or reduced to dead ash.

Our ancestors understood soil not as an inert factory floor, but as a breathing, living mother. When we set fire to crop residues, the intense surface heat incinerates the topsoil microbiome: earthworms, beneficial mycorrhizal fungi, and nitrogen-fixing bacteria are obliterated in minutes.

"मृदा एव जीवनम् — *Soil is Life itself.*"

— Ancient Vedic Agricultural Maxim

Consequently, Soil Organic Carbon (SOC) has plummeted below **0.5%** in over 65% of Indian agricultural land, compared to a healthy baseline of 1.5%–2.0%. Composting is our collective, biological path back to living soil.

Losses per Tonne of Straw Burned

Nitrogen (N) Lost:	5.5 kg / tonne
Phosphorus (P) Lost:	2.3 kg / tonne
Potassium (K) Lost:	25 kg / tonne
Organic Carbon Lost:	400 kg / tonne
Soil Microbial Kill Rate:	Top 2.5 inches sterilized



The Two Paths: Burning vs Composting

A stark economic and biological comparison for every farming household



THE DESTRUCTIVE BURNING CYCLE

Short-term speed at extreme long-term cost

- 🌾 **₹8,000 / acre lost yearly** in destroyed N, P, K and micro-nutrients.
- 🌾 **100% Surface Microbe Death:** Earthworms, fungi, and azotobacter incinerated.
- 🌾 **Severe Air Smog:** Emission of PM2.5, carbon monoxide, and toxic aerosols.
- 🌾 **Soil Hard-Pans:** Soil becomes compacted, losing porosity and water-holding ability.
- 🌾 **Chemical Dependency:** Farmer forced to buy more synthetic urea & DAP every season.

OUTCOME: DEPLETED SOIL · RISING DEBT · TOXIC AIR



THE REGENERATIVE COMPOSTING CYCLE

Harnessing biology to build lasting generational wealth

- 🌾 **₹8,000 / acre input savings** by replacing chemical fertilizers with rich compost.
- 🌾 **Microbial Explosion:** Fosters billions of beneficial bacteria and native earthworms.
- 🌾 **Zero-Emission Air:** Converts raw biomass into stable, smell-free soil carbon.
- 🌾 **Moisture Sponge:** Improves irrigation retention by 20–30%, saving groundwater.
- 🌾 **Humus Creation:** Restores deep golden fertility that lasts for multi-crop cycles.

OUTCOME: SOIL GOLD · HIGH YIELD RESILIENCE · CLEAN AIR

CHAPTER 02

The Carbon-Nitrogen Equation

 *The chemical stoichiometry of rapid, odor-free decomposition*

Composting is not random rotting—it is a biological furnace driven by billions of microorganisms. These microscopic workers operate under a strict nutritional rule known as the **Carbon-to-Nitrogen (C:N) Ratio**.

Microbes use **Carbon (C)** as their primary energy fuel (their food carbohydrates) and **Nitrogen (N)** to synthesize amino acids, cellular proteins, and enzymes for rapid reproduction.

To achieve the fastest, highest-temperature decomposition without unpleasant smells, the ideal starting recipe requires a balanced ratio of approximately **30:1** (30 parts Carbon for every 1 part Nitrogen).

 **PRO TIP:** Avoid adding eucalyptus leaves, plastic, glass, battery acid, or foliage treated with synthetic herbicides (like 2,4-D or Glyphosate), as they will poison the microbial engine.

Categorizing Farm Materials

✓ **BROWNS (Carbon-Rich / Energy Source):**

Wheat straw, paddy stubble, sugarcane bagasse, dry deciduous leaves, shredded cardboard, dry corn stalks.

✓ **GREENS (Nitrogen-Rich / Protein Source):**

Fresh cattle dung, green weed trimmings, legume/pulse haulms, kitchen vegetable peels, poultry manure.

✓ **ACTIVATORS (Microbial Inoculants):**

Rich virgin forest soil, aged compost slurry, or fresh cow urine-jaggery solution (Jeevamrit) to seed indigenous microbes.

What happens when unbalanced?

- *Too Much Carbon (>50:1):* Pile stays cold and decomposition stalls for months.
- *Too Much Nitrogen (<15:1):* Pile turns anaerobic, smells foul of ammonia, and loses nitrogen gas.

Common Farm Materials & Their C:N Ratios

Mastering the art of combining high-carbon straws with nitrogenous manures

Material Name	Category	C:N Ratio	Function in Pile
Wheat Straw	Brown	80 : 1	Structure & high carbon energy
Rice / Paddy Straw	Brown	65 : 1	Carbon reservoir; needs fine chopping
Dry Forest Leaves	Brown	60 : 1	Mineral-rich carbon bulk
Fresh Cattle Dung	Green	20 : 1	Moisture & bacterial inoculant
Fresh Grass / Weeds	Green	15 : 1	Rapid nitrogen fuel for heat
Kitchen Vegetable Waste	Green	15 : 1	Fast nutrient and water release
Green Legume Haulms	Green	12 : 1	Potent organic nitrogen booster

THE GOLDEN COMBINATION RULE:

2-3 BASKETS BROWNS (Carbon)

1 BASKET GREENS (Nitrogen)

Balancing Equations in the Village

Most Indian farmers have an abundance of high-carbon crop residues (such as wheat stubble at 80:1 or paddy straw at 65:1). Because these have too much carbon, they sit in the field without breaking down.

- Step 1:** Chop tough straw into 2-4 inch lengths to give bacteria 10× more surface area.
- Step 2:** Mix thoroughly with 30-40% fresh cattle dung slurry or green cover crop cuttings.
- Step 3:** Water evenly until the combined ratio naturally settles at the biological optimum of 30:1.



CHAPTER 03

Building the Aerated Pile

 *Constructing a self-insulating, breathing aerobic bio-reactor*

A compost pile is a living biological structure. It must breathe constantly. Without adequate oxygen, anaerobic bacteria take control, dropping the internal temperature, producing foul sewer gases (hydrogen sulfide), and producing low-grade, acidic inputs.

The Golden Dimension Rule: Select a well-drained, shaded site on your farm. Build a pile that is strictly **4 feet wide, 4 feet high**, and as long as needed.

Piles smaller than 3 feet cube lose heat too rapidly to the atmosphere and cannot reach pasteurization temperature. Piles wider than 5 feet suffocate in the center due to lack of airflow.

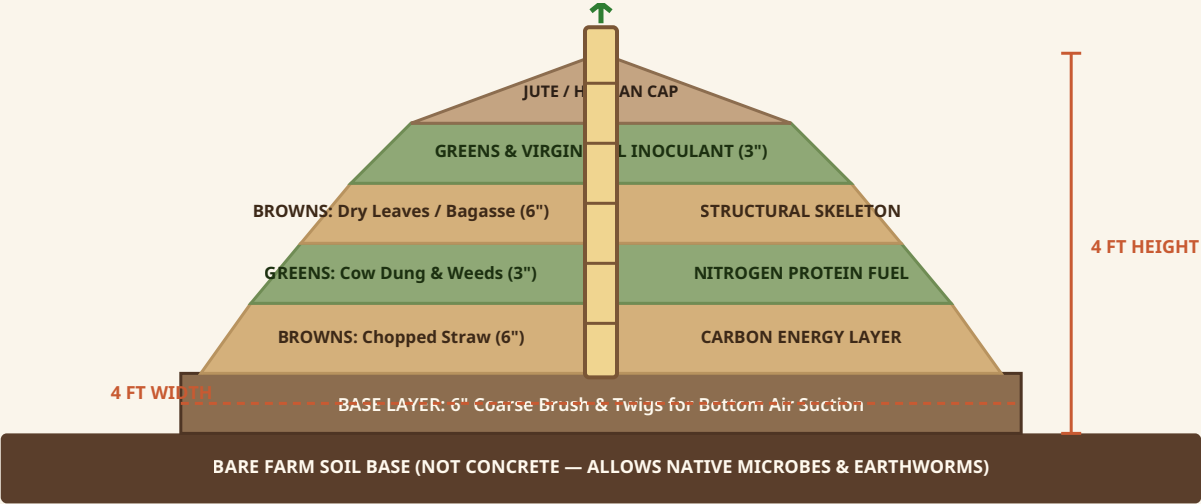
 **PRO TIP (Passive Bamboo Chimney):** Place a 5-foot bamboo pole with 1-inch holes drilled every 6 inches vertically in the center of the pile. This acts as a perpetual natural draft chimney, drawing fresh air into the deepest core.

Step-by-Step Stacking Protocol

- 1. Aeration Base Layer (6 inches):**
Lay coarse tree twigs, pigeon pea stalks, or dry woody brush directly on bare ground to ensure bottom drainage and suction.
- 2. Brown Carbon Layer (6 inches):**
Add chopped straw or dry leaves. Moisten with water.
- 3. Green Nitrogen Layer (2–3 inches):**
Spread fresh cattle dung, green weeds, or kitchen scraps.
- 4. Microbial Dusting (1 inch):**
Sprinkle mature forest soil or aged compost to inoculate local fungi.
- 5. The Squeeze Test:**
Ensure the pile feels like a *wrung-out damp sponge* (50–60% moisture). Squeezing a handful should produce only 1–2 drops of water.

Cross-Section Architecture of the 4×4 Pile

Visualizing airflow, internal stratification, and insulation geometry



CHAPTER 04

The Fire Within

🔥 *Biological pasteurization and thermal disease sterilization*

Within 48 to 72 hours of stacking an aerated pile with the correct 30:1 ratio, an extraordinary phenomenon occurs: the internal temperature climbs steeply from ambient air level to **55°C–65°C (131°F–149°F)**.

This intense heat is not produced by solar radiation. It is the biological heat generated by trillions of thermophilic (heat-loving) bacteria rapidly devouring simple carbohydrates and cellulose.

Why the High Temperature is Essential: Sustained temperatures above 55°C for 5–7 consecutive days effectively pasteurize the material. It kills pathogenic microorganisms (such as *E. coli* and *Salmonella*), sterilizes troublesome weed seeds (like *Phalaris minor* and *Parthenium*), and destroys fungal spores.

"The heat of the compost is the flame of microbial life."

🪵 The Traditional Wooden Stick Test

If you do not own a long compost probe thermometer, use this traditional village field technique:

1. Push a dry, smooth neem or bamboo stick 3 feet deep into the pile's center.
2. Leave it undisturbed for 30 minutes.
3. Pull the stick out and immediately grasp the tip firmly in your palm.

Hot & Steaming (55°C–65°C): Perfect! Thermophilic bacteria are thriving.

Warm & Damp (40°C–50°C): Good, but turning may be needed soon to boost oxygen.

Cold Stick (<35°C): The pile is either too dry (needs water) or deficient in nitrogen (needs cow dung slurry).



Thermophilic Bacteria

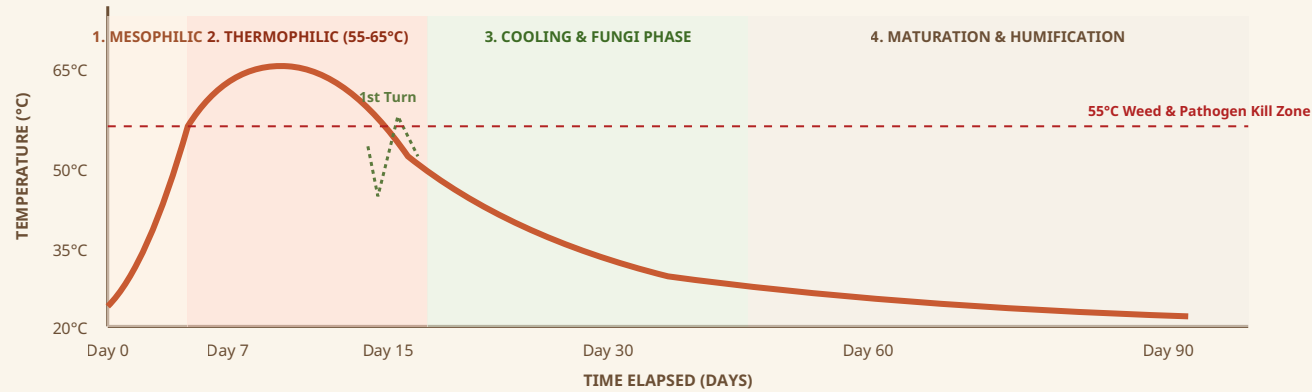
Bacillus & Actinobacteria colonies



Active Enzyme

The 90-Day Biological Temperature Curve

Tracing the four biological phases of aerobic decomposition



Key Takeaway: Notice how temperatures spike dramatically in the first 7 days, pasteurizing the mass, before gradually descending into the cooling and curing phases where actinomycetes and earthworms assemble complex humus molecules.

CHAPTER 05

Turning & Aerating

 *Re-energizing the microbial engine with fresh oxygen*

As thermophilic microbes rapidly consume carbon, they deplete the oxygen within the pile's pore spaces. When oxygen drops below 5%, aerobic microbes go dormant, temperatures decline, and decomposition slows to a halt.

To reignite the thermal engine, the entire pile must be **turned inside out**. This physical inversion serves three vital goals:

-  **Restores 21% Atmospheric Oxygen:** Replenishes fresh air deep in the core.
-  **Uniform Pasteurization:** Cool outer crust materials are moved into the hot center.
-  **Particle Breakage:** Distributes fungal colonies throughout the mass.

Turn the pile every **10 to 14 days** for the first month, and once every 20 days thereafter.

 12-Week Transformation Schedule

WEEKS 1–2 (Active Heat Peak):
Pile is fiercely hot (55–65°C), shrinks by 30% in volume, emits sweet vapor.

WEEKS 3–4 (First & Second Turn):
Turn pile thoroughly. Reheat occurs within 24 hours. White actinomycete filaments appear.

WEEKS 5–8 (Cooling & Fungi Stage):
Temperature settles to 35–40°C. Earthworms begin colonizing outer margins. Material darkens.

WEEKS 9–12 (Curing & Humification):
Pile cools to ambient temperature. Smells of fresh petrichor. Fully stabilized soil gold.

The 4-Step Inside-Out Turning Method

How to properly invert and aerate your farm compost pile with simple hand tools

1 Peel the Outer Shell

Using a 4-prong pitchfork, skim off the dry outer 4–6 inches of the pile. Set this unfermented dry material aside in a separate starter pile next to the original site.

2 Move Core Material Outward

Fork the hot, dark, steaming center core and place it along the outer perimeter of your new pile location. This ensures the already pasteurized core gives way to unfinished material.

3 Invert Outer Shell to Deep Center

Take the dry outer material you set aside in Step 1 and bury it directly into the new center. Sprinkle lightly with cow dung slurry to kickstart immediate re-heating.

4 Moisture Adjustment & Fluffing

As you reconstruct layers, sprinkle water with a rose can if the straw feels dry. Do not compact or step on the pile; leave it loose and fluffy to maximize oxygen retention.



CHAPTER 06

Recognizing Soil Gold

✨ *Precise sensory and scientific tests to verify compost readiness*

After 60 to 90 days of disciplined aerobic management, the pile's internal temperature will naturally stabilize at the ambient air temperature. This indicates that the readily degradable carbon has been transformed into stable **humus**.

True finished compost—what our elders called *Mrid-Swarna* (Soil Gold)—possesses unmistakable physical characteristics:

- 🌿 **Color:** Deep chocolate brown to jet black.
- 🌿 **Texture:** Light, fluffy, crumbly like moist chocolate cake.
- 🌿 **Aroma:** Smells cleanly of forest floor after the first monsoon shower (caused by *geosmin* produced by actinomycetes).

⚠️ **DANGER OF UNFINISHED COMPOST:** Applying raw, immature compost to crops locks up available soil nitrogen (nitrogen robbery), starving young seedlings and burning root tips.

🔪 The Sealed Plastic Bag Maturity Test

A simple, foolproof village test to confirm your compost is 100% finished and safe:

1. Take a handful of moist compost from the pile's center.
2. Seal it tightly inside an airtight transparent plastic zip-lock bag.
3. Leave the bag at room temperature in a dark room for **48 hours**.
4. Unzip the bag and take an immediate deep sniff.

Sweet, Earthy Forest Smell: Compost is mature, stabilized, and ready for field broadcast.

Sour, Ammonia, or Rotten Odor: Microbes are still consuming raw carbon. Return to pile and cure for 2 more weeks.


"Mrid-Swarna · Pure Soil Gold"

Diagnostic Checklist: Ready vs Unready Compost

Evaluate your compost heap against these strict agronomic criteria before field spreading



UNFINISHED COMPOST (DO NOT APPLY)

Temperature:	Warm or hot to the touch (>35°C).
Visual:	Individual straw pieces, leaves, or dung balls clearly recognizable.
Aroma:	Sharp ammonia, vinegar, sour cabbage, or foul rotting odor.
Texture:	Chunky, stringy, matted, sticky, or slimy when compressed.
Crop Effect:	Locks up soil nitrogen, rots seeds, attracts maggot flies.



MATURE SOIL GOLD (READY FOR SOWING)

Temperature:	Completely cool (ambient air temperature).
Visual:	Homogeneous dark brown/black crumbly humus. Original raw matter fully disappeared.
Aroma:	Rich, soothing scent of forest petrichor and damp virgin soil.
Texture:	Soft, spongy, loose particles that crumble freely through fingers.
Crop Effect:	Instantly feeds root zone, stimulates mycorrhizae, boosts yield.

CHAPTER 07

Bringing Carbon to the Field

 *Agronomic guidelines for application dosage, timing, and soil placement*

Compost is not a chemical shot of synthetic fertilizer. It is a long-term biological overhaul of your soil's physical architecture, moisture retention sponge, and Cation Exchange Capacity (CEC).

Timing & Incorporation: For field crops, broadcast compost across the field **2 to 3 weeks prior to sowing**. Till gently into the top 4–6 inches (the rhizosphere zone) where root systems develop. Avoid burying compost deep beneath the furrow slice where anaerobic conditions prevent respiration.

Orchard Ring Technique: For mango, guava, or citrus orchards, dig a shallow ring trench around the drip-line of the tree canopy. Fill with 10 kg compost and cover with dry leaf mulch.

 Recommended Dosage by Crop Category

Crop Type	Dosage / Acre	Application Method
Cereals (Wheat, Paddy)	2.5 – 3.0 Tonnes	Broadcast before final harrowing
Cash Crops (Cotton, Cane)	4.0 – 5.0 Tonnes	Band placement along planting furrows
High-Value Vegetables	4.0 – 5.0 Tonnes	2-inch raised bed topsoil integration
Fruit Orchards	10 – 15 kg / tree	Canopy dripline shallow ring trench
Reclaiming Degraded Soils	6.0 – 8.0 Tonnes	Split dosage: 50% summer, 50% winter

Water Savings Benefit: Soils enriched with 3 tonnes/acre of organic compost retain up to **100,000 liters** more rainwater per acre, shielding crops from drought dry-spells.

FINAL CHAPTER & EPILOGUE

The Circle of Return

 *Restoring the sacred covenant between the Indian farmer and Mother Earth*

True farming is the sacred practice of returning what we borrow. The crop draws minerals, water, and microbial energy from the soil. When harvest comes, the grain sustains human society, while the straw and dung are returned to the compost pile.

Through the alchemy of microbial decomposition, this carbon returns home to nourish future generations.

By abandoning the torch of residue burning and taking up the pitchfork of aerobic composting, every Indian farmstead can save upwards of **₹8,000 per acre** annually, eradicate chemical dependency, revitalize groundwater tables, and restore pride in our agricultural heritage.

"जैविक खेती ही सुखी खेती है"

Organic farming is the path to happy, resilient, and prosperous farming.



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